**CodeAlpha IoT Internship**

**Task 2 – Sensor-Based Simulation**

**Project Title**

**Smart Home Temperature Monitoring and Automatic Cooling System**

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## 1. Introduction

The Internet of Things (IoT) has transformed the way modern homes operate by connecting sensors, controllers, and smart devices through communication networks. Smart home systems improve comfort, security, and energy efficiency by automating daily activities based on real-time environmental conditions.

This project demonstrates a Smart Home Temperature Monitoring and Automatic Cooling System using an Arduino Uno and a TMP36 temperature sensor. The system continuously monitors room temperature and automatically alerts users whenever the temperature exceeds a predefined threshold. The project was developed and tested using the Tinkercad Circuits simulation platform without requiring physical hardware.

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## 2. Project Objective

The primary objective of this project is to design and simulate an intelligent temperature monitoring system capable of:

- Monitoring room temperature continuously.
- Displaying live temperature readings on an LCD.
- Detecting high-temperature conditions.
- Providing both visual and audible alerts.
- Demonstrating the practical implementation of IoT concepts through simulation.

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## 3. Components Used

The following components were used in this project:

- Arduino Uno R3
- TMP36 Temperature Sensor
- 16×2 LCD Display
- Green LED
- Red LED
- Piezo Buzzer
- Breadboard
- 220Ω Resistors
- 10kΩ Potentiometer
- Jumper Wires

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**4. System Architecture**

The project follows the standard three-layer IoT architecture.

**Sensing Layer**

The TMP36 temperature sensor continuously measures the ambient temperature and converts it into an analog voltage signal.

**Processing Layer**

The Arduino Uno reads the analog voltage using its built-in Analog-to-Digital Converter (ADC). The temperature is calculated using the following equation:

**Temperature (°C) = (Voltage – 0.5) / 0.01**

The Arduino then compares the measured temperature with the predefined threshold value of **30°C**.

**Output Layer**

The processed information is presented to the user through:

- LCD display
- Green LED
- Red LED

- Piezo Buzzer

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**5. Project Workflow**

The system operates according to the following sequence:

1. Read temperature from the TMP36 sensor.
2. Convert analog voltage into degrees Celsius.
3. Display the current temperature on the LCD.
4. Compare the reading with the threshold value.
5. Activate the appropriate outputs.
6. Repeat the process every second.

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**6. System Operation**

**Normal Mode (Temperature < 30°C)**

When the measured temperature is below 30°C:

- Green LED turns ON.
- Red LED remains OFF.
- Buzzer remains OFF.
- LCD displays:

Temperature: XX.X°C

Status: Normal

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**Alert Mode (Temperature ≥ 30°C)**

When the temperature reaches or exceeds 30°C:

- Green LED turns OFF.
- Red LED turns ON.
- Piezo buzzer produces a 1000 Hz warning tone.

- LCD displays:

Temperature: XX.X°C

Status: HIGH TEMP!

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**7. Technical Features**

This project includes several practical Arduino programming techniques:

- Continuous analog sensor monitoring
- Real-time LCD updates
- 4-bit LCD communication mode
- Threshold-based decision making
- Degree symbol display
- LED status indicators
- Audible alarm using the tone() function
- Serial Monitor support for debugging

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**8. Results**

The simulation successfully demonstrated automatic temperature monitoring and alert generation.

The system accurately distinguished between normal and high-temperature conditions and provided immediate feedback through visual indicators, LCD messages, and audible alarms.

Testing within Tinkercad confirmed that the system responded correctly whenever the temperature crossed the predefined threshold.

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**9. Future Improvements**

Several enhancements can be implemented in future versions of the project, including:

- Wi-Fi connectivity using ESP8266 or ESP32.

- Cloud-based temperature monitoring.
- Mobile application integration.
- Email or SMS notifications.
- Data logging using cloud databases.
- Automatic fan control using a relay module.
- Integration with smart home platforms.

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### 10. Conclusion

This project successfully demonstrates the fundamental concepts of the Internet of Things using a smart home temperature monitoring application. Through the integration of sensing, processing, and output devices, the system continuously monitors environmental conditions and responds automatically when abnormal temperatures are detected.

Although implemented as a simulation, the project closely represents how real-world smart home monitoring systems operate. It also highlights the importance of automation, real-time monitoring, and intelligent decision-making in modern IoT applications.

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### References


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